LESSON 3.1
EQUIVALENT CONVERSIONS: FRACTIONS DECIMALS; DECIMALS PERCENTAGES

LESSON INTRODUCTION

MA.6.NSO.3.5 Rewrite positive rational numbers in different but equivalent forms including fractions, terminating decimals, and percentages.

LEARNING TARGETS

- I can convert between fractions and decimals.
- I can convert between percentages and decimals.

BENCHMARK COHERENCE

BUILDING ON

MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction.

WORKING TOWARD

MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.

ESSENTIAL QUESTIONS

- How can you use place value to convert a decimal to a fraction?
- How do you convert a percentage to a decimal?
- What are the similarities and differences between converting fractions, decimals, and percentages?

LESSON NARRATIVE

In this lesson, students learn how to convert between fractions and decimals and between percentages and decimals. 3.1.1 Warm-Up activates prior knowledge of place values by analyzing the different numerical values of digits to determine place value. Through Guided Instruction, students make connections between place values in decimals and denominators in fractions as they learn to convert between fractions and decimals, and between decimals and percentages.

COMPONENT SUMMARY

3.1.1 WARM-UP *(~5 MIN.)*

Self-evaluation on place value

3.1.2 GUIDED INSTRUCTION *(15-20 MIN.)*

Explore relationships of value, decimals, and fractions

3.1.3 GUIDED PRACTICE *(5-10 MIN.)*

Practice converting between fractions and decimals

3.1.4 GUIDED INSTRUCTION *(5-10 MIN.)*

Making connections among percentages, decimals, and fractions

3.1.5 GUIDED PRACTICE *(10 MIN.)*

Practice converting between percentages and decimals

3.1.6 YOUR TURN *(5 MIN.)*

Practice converting between percentages, decimals, and fractions

3.1.7 WRAP-UP *(~5 MIN.)*

Which one does not belong activity

RESOURCES

GLOSSARY

Key Terms:

- Percentage

Additional Terms:

- Place value

MATERIALS

- (Optional) Place value chart
- (Optional) Chart paper

ADDITIONAL PREPARATION

- (Optional) Create and laminate a decimal place value chart for students to use or reference throughout the lesson and activities.
- (Optional) Create an anchor chart that can be used in 3.1.2 Guided Instruction

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may put the decimal point in the wrong place.
- Students may not know where to put the decimal point on a whole number.
- Students may mix up the numerator with the denominator.
- Students may read decimals incorrectly (0.625 as 625 hundredths instead of 625 thousandths)

THINGS TO CONSIDER

- Use student responses from the self-evaluation in 3.1.1 Warm-Up to possibly review the rules for dividing and multiplying decimals and to gauge scaffolding needed for subsequent lessons in this unit related to prior knowledge on fractions and decimals.
- When questioning and modeling problems, display and encourage the use of accurate math vocabulary including percentage, decimal, fraction, and place value to help students make connections and notice differences.

LITERACY AND LANGUAGE ROUTINES

Take Turns

3.1.3 and 3.1.5 Guided Practices provide opportunities for students to explain, justify, agree, or disagree with conversion problems. Consider modeling "Take Turns" routines for the first problems in each section to help students identify and explain the relationship between place value, decimals, fractions, and percentages.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Students will be working to convert between decimals, fractions, and percentages. Appropriate and efficient methods are necessary in finding accurate answers and building fluency in recognizing different equivalent forms. Offer students flexibility in converting numbers, while providing opportunities for math-rich discussions about different possible methods.

DIFFERENTIATE INSTRUCTION

For visual learners, consider providing struggling students with a laminated decimal place value chart to help them when converting with decimals. Consider incorporating a gallery walk for kinesthetic learners.

3.1.1 WARM-UP: SELF-EVALUATION

Component Overview: *Students individually rate their confidence on a series of different conversations from decimals, fractions, and/or percentages. Consider using the information provided by this evaluation to assess the amount of scaffolding required in subsequent lessons and throughout this unit.*

DIFFERENTIATED INSTRUCTION

Consider reviewing the confidence scale results to determine the level of scaffolding, if any, needed during activities in this lesson and subsequent lessons in this unit. Student responses may also be helpful in creating optional groups later in the lesson during Guided Practice sections.

3.1.1 Warm-Up: Self-Evaluation

1. For each problem, rate your level of confidence in your ability to answer each question, but don't actually answer them.

   Use a scale from 1 to 5, where 1 means you do not feel confident about how to answer the question, and 5 means you feel very confident.

Problem	Confidence Level
Rewrite $2\frac{2}{5}$ in its equivalent decimal form.	*Answers will vary.*
Rewrite 0.625 as a fraction with the same value.	*Answers will vary.*
Rewrite 0.9 as an equivalent percentage.	*Answers will vary.*
Rewrite 75% in its equivalent decimal form.	*Answers will vary.*

3.1.2 Guided Instruction: Fraction and Decimal Conversions

Decimal Place Value Chart

Thousands	Hundreds	Tens	Ones	Decimal Point	Tenths	Hundredths	Thousandths

The place value of a number provides the numerical value that a digit has by virtue of its position in a number.

Example: In 278.493
the 8 is in the **ones** place
the 4 is in the **tenths** place
the 9 is in the **hundredths** place
the 3 is in the **thousandths** place

1. **Use the original number to fill in the missing digits and place values.**

Original Number	Digit	Place Value
422.54	5	Tenths
5.276	5	Ones
12.749	4	*Hundredths*
3.826	6	Thousandths

2. **Consider the number 0.08.**
   a. **In the number 0.08, what is the place value of the final digit?**
   Hundredths

   b. **What is the written description of the fraction $\frac{8}{100}$?**
   8 *hundredths*

   c. **Discuss with your partner any similarities or differences between the decimal 0.08 and the fraction $\frac{8}{100}$.**

When rewriting a decimal that is greater than 0 but less than 1 as an equivalent fraction, use the place value of the final digit to determine the denominator of the fraction.

3. **In the decimal 0.7, the final digit is in the**
   - o tenths
   - o hundredths
   - o thousandths

   place, so
   - o $\frac{7}{10}$
   - o $\frac{7}{100}$
   - o $\frac{7}{1000}$

   is the equivalent fraction form.

4. **In the decimal 0.037, the final digit is in the**
   - o tenths
   - o hundredths
   - o thousandths

   place, so
   - o $\frac{37}{10}$
   - o $\frac{37}{100}$
   - o $\frac{37}{1000}$

   is the equivalent fraction form.

3.1.2 GUIDED INSTRUCTION: FRACTION AND DECIMAL CONVERSIONS

Component Overview: *Students activate prior knowledge of place values by using a decimal place value chart to understand the value of digits in different place values. Students then make connections between place values in decimals and denominators in fractions as they learn to convert between fractions and decimals.*

DIFFERENTIATED INSTRUCTION

Consider displaying or having students use a laminated decimals place value chart to reference place value and decimals. This entry point for visual learners can be used in subsequent activities and lessons in which decimals conversions are practiced.

ENRICHMENT

Notice the question prompts in questions 2a-c. Position students to discover the relationship between the place value and the fraction equivalent by exploring questions 2a-c with a partner.

ENRICHMENT

Monitor student responses and listen for opportunities to extend their thinking.

- "The decimal and fraction are the same." What do you mean by the same? Which part is the same? What do we call that in math? (equivalent)
- "They both have two zeros." Will that always be true? Which part of the expression determines 'how many zeros' or the place value of the fraction? Does that work for this example?

QUESTIONING

Are your students ready for a challenge? Consider briefly incorporating a number like 5.45 or 109.032 to see what students do with the 5 and the 109. Some may omit the values, while others may place them in the numerator, while others may represent the expression as a mixed number.

3.1.2 GUIDED INSTRUCTION: FRACTION AND DECIMAL CONVERSIONS (CONTINUED)

DIFFERENTIATED INSTRUCTION

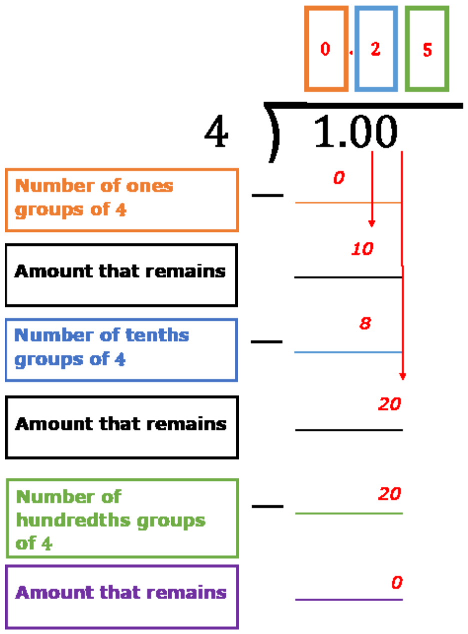
Consider an anchor chart for long division that highlights conceptual understanding while supporting a variety of entry points.

ENRICHMENT

Why do we need to write the decimal to the right of the whole number dividend in this problem? Why is it important to write the decimal in the quotient lined up with the same place of the dividend?

A fraction can be converted into its equivalent decimal form by dividing the ***numerator*** by the ***denominator***.

5. To convert the fraction $\frac{1}{4}$ to a decimal, divide 1 by 4 by completing the diagram below.



3.1.3 Guided Practice: Fraction and Decimal Conversions

1. Complete the table by converting each decimal to an equivalent fraction. If possible, write the equivalent fraction so that the numerator and denominator have no common factors.

Decimal Form	Fraction Form
0.08	$\frac{2}{25}$
0.27	$\frac{27}{100}$
2.2	$2\frac{2}{10} = 2\frac{1}{5}$
0.005	$\frac{5}{1000} = \frac{1}{200}$

2. Complete the table by rewriting each fraction as its equivalent decimal.

Fraction Form	Your Work	Decimal Form
$\frac{1}{5}$	*Answers will vary.*	0.2
$\frac{5}{8}$	*Answers will vary.*	0.625
$\frac{7}{200}$	*Answers will vary.*	0.035

3.1.3 GUIDED PRACTICE: FRACTION AND DECIMAL CONVERSIONS

Component Overview: *Students convert between equivalent fraction and decimal forms.*

TEACHER MOVES

Take Turns

Use Take Turns language routines for question 1 by providing opportunities for students to explain, justify, agree, or disagree with answers given by peers. Support student discussions with sentence stems, including:

- I know ___ is equivalent to ___ because…
- I disagree because…
- I agree because…

TEACHER MOVES

Monitor student responses while they are practicing. Common misconceptions could include:

- Misplacing the decimal (2.0 instead of 0.2)
- Mixing up the numerator and denominator
- Reading decimals (0.625 as 625 hundredths instead of 625 thousandths)

3.1.4 GUIDED INSTRUCTION: DECIMAL AND PERCENTAGE CONVERSIONS

Component Overview: *Students use knowledge of fractions and decimals to determine the connection to percentages. The definition and notation of percentages are introduced and students learn how to convert between decimals and percentages.*

DIFFERENTIATED INSTRUCTION

To strengthen student understanding of percentage as a ratio out of 100, ask students to think of words they know that contain "cent," meaning 100, then have them explain how the word relates to 100. Examples include:

- A cent = 1/100th of a dollar; 100 cents in a dollar
- Century = 100 years
- Centimeter = 1/100th of a meter; 100 centimeters in a meter
- Centipede = a bug with 100 legs

3.1.4 Guided Instruction: Decimal and Percentage Conversions

A ***percentage*** is a ratio expressed as a fraction with a denominator of 100.

Percentages can also be written as fractions with a denominator of 100, or as a decimal.

1. Write 66% as a fraction and as a decimal.

Percent	Fraction	Decimal
66%	$\frac{66}{100}$	0.66

To convert a decimal to a percent, multiply the decimal by 100.

2. Convert the decimal 0.16 to a percent.

$$0.16 \times \underline{\ \ 100\ \ }$$
$$\underline{16\%}$$

To convert a percent to a decimal, divide the percent by 100 or multiply by $\frac{1}{100}$.

3. Convert 16.2% to a decimal.

$$16.2 \div \underline{\ \ 100\ \ } \text{ or } 16.2 \cdot \frac{1}{100}$$
$$\underline{0.162}$$

3.1.5 Guided Practice: Decimal and Percentage Conversions

1. Complete the table by converting each decimal to an equivalent percentage.

Decimal Form	Percent Form
0.08	8%
0.27	27%
2.2	220%
0.005	0.5%

2. Rewrite each percentage into its equivalent decimal form to complete the table.

Percent Form	Decimal Form
72%	0.72
12.5%	0.125
0.7%	0.007
214%	2.14

3.1.5 GUIDED PRACTICE: DECIMAL AND PERCENTAGE CONVERSIONS

Component Overview: *Students convert between different equivalent decimal and percentage forms.*

ENRICHMENT

For question 1, what do you notice about the percentage conversions for 2.2 and 0.005? Why is 2.2 written as a percent larger than 100? Why is 0.005 written as a percent smaller than 1%, even though the number 5 is greater than the number 2?

TEACHER MOVES

Consider having students engage in Take Turns language routines by verbally explaining how they converted the percentages to decimal form in question 2. This can position students to explain, justify, agree, or disagree with answers provided by peers.

3.1.6 YOUR TURN!

Component Overview: *Students practice converting between decimals, fractions, and percentages with multiple problems. Consider having students complete some or all the problems based on available time and necessary practice for the different representations.*

ENRICHMENT

One of the most common misconceptions finds students writing single digit percentages as tenths, like question 1b students writing 3% as 0.3. Consider asking probing questions that position students to connect percent as “out of 100” in order to conceptualize 3% = 3/100 = three hundredths = 0.03.

DIFFERENTIATED INSTRUCTION

Consider using graph paper or base-10 blocks as visual scaffolding for aligning and categorizing place values and decimals.

3.1.6 Your Turn!

1. **Rewrite each percentage as a decimal.**

   a. 125.6% 1.256 b. 3% 0.03

2. **Convert each decimal to an equivalent percentage.**

   a. 0.008 0.8% b. 4.71 471%

3. **Rewrite each fraction as a decimal.**

   a. $\frac{1}{2}$ 0.5 b. $2\frac{1}{10}$ 2.1

4. **Rewrite each decimal as an equivalent fraction. Simplify where applicable.**

   a. 0.94 $\frac{94}{100} = \frac{47}{50}$ b. 1.7 $1\frac{7}{10}$

3.1.7 Wrap-Up: Ticket Out the Door

1. Circle the expression that does not belong.

$\frac{2}{5}$	0.4
40%	$\frac{4}{8}$

2. Explain your reasoning for your choice.
Answers will vary. $\frac{4}{8}$ is equivalent to $\frac{1}{2}$ which is written as 0.5 as a decimal and 50% as a percentage. $\frac{2}{5}$, 0.4, and 40% are all equivalent.

3.1.7 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *This Wrap-Up assesses students' ability to convert between fractions, decimals, and percentages and determine which expressions are equal by identifying which representation does not belong. Consider providing written feedback to each student on their explanation and include one point of positive feedback to the student.*

DIFFERENTIATED INSTRUCTION

Consider how differing responses can tell you different elements of students' learning progression.

- Based on the learning targets for this lesson, most students might choose 4/8 to correctly identify as the only expression not equivalent.
- Students may also choose the decimal or percent as the only decimal or percent shown, while other students may identify 2/5 as the only option without a "4" in it, which would be a good opportunity to probe students on equivalence and multiple representations.
- A "correct" response should not be the focus as much as the student using appropriate mathematics language in their reasoning.